

Blake Allen Linders

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Education

Texas A&M University

Bachelor of Science in Electrical Engineering, Minor in Math – GPA: 3.51

August 2024 – Present

(Expected May 2028)

- **Relevant Coursework:** Electrical Circuit Theory, Digital Systems Design, Machine Learning for EE, Program Design & Concepts, Electricity & Magnetism, Differential Equations, Applied Mathematics
- Data Analysis Certificate – Katzberg Institute
- IEEE Member and Multiple Intramural Sports

Skills

Hardware/Lab Tools: FPGA, Oscilloscope, DAQ, Multimeters, Breadboard

Software/Simulation: Vivado, LTspice, MATLAB, Simulink, Tableau, Git/GitHub

Programming Languages: Python (NumPy, SymPy, pandas, Plotly, Matplotlib, Scikit-learn, TensorFlow), PyTorch, C++, Verilog (Behavioral & Structural), SQL, Java, JavaScript, HTML, CSS

Engineering Concepts: Machine Learning, Signal Processing, Data Analysis, Embedded Systems, Digital Systems Design, Finite State Machines, Circuit Theory, Op-Amps, Logic Gates, REST APIs

Projects

Three Wheel ESP32 RC Car — *Independent Project*

May 2026

- Designed a 3.3V/5V mixed-logic hardware architecture around an ESP32 and L293D motor driver, engineering a custom resistor voltage divider to perform 5V-to-3.3V logic-level shifting for safe integration of an ultrasonic collision sensor.
- Developed a low-latency WebSockets control system hosted on an ESP32 SoftAP with a DNS captive portal, enabling simultaneous multi-touch processing from a custom HTML/JS interface to drive hardware PWM signals with a 500ms dead-man's safety switch.

Machine Learning & Data Analysis — *Academic Project*

January 2026 - May 2026

- Analyzed complex neural network architectures, calculating VGG16 CNN weights and mapping decision tree boundaries to evaluate predictive model performance.
- Executed K-means clustering iterations using Python and utilized Business Intelligence frameworks to extract, process, and analyze actionable insights from raw data.

FPGA Digital Systems & Hardware Validation — *Academic Project*

August 2025 - December 2025

- Engineered multi-state FSMs using Verilog to manage complex timing for a multi-directional traffic intersection and validate sequential multi-bit tactile button inputs.
- Synthesized logic via Vivado and deployed state machines to a ZYBO Z7-10 board, integrating clock dividers to translate high-frequency system clocks into real-time physical delays.
- Mapped FPGA signals to a breadboard to drive LED indicators and characterized external components by measuring I/V curves with an oscilloscope to optimize empirical voltage drops.

Experience & Leadership

Undergraduate Studies Internship Program

May 2026 – August 2026

Coding for Data Academic Intern

- Completed an academic internship focused on programmatic problem-solving, applying statistical filtering, multi-variable aggregations, and algorithmic data cleaning to solve real-world computational challenges across diverse industry case studies.
- Engineered complex SQL pipelines—utilizing advanced joins, subqueries, and conditional logic—to model Intel hardware lifecycles, integrating large-scale device data to quantify CO2 reductions and optimize e-waste repurposing strategies.
- Developed Python data acquisition and visualization workflows using Pandas and Plotly to analyze complex systems, from planetary meteorological models to web traffic analytics for The Grammys, translating raw datasets into structured performance metrics.

Blue Stars Drum and Bugle Corps

May 2025 – August 2025

Section Leader (Bassline)

- Outstanding First Year Battery Member Award.

Lewisville High School

August 2024 – April 2025

Drumline Technician

Pacific Crest Youth Arts Organization

May 2024 – August 2024

Performing Member (Bassline)